

Candidate Name

Centre Number

Candidate Number



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Ordinary Level

PHYSICS

PAPER 3 Practical Test

4023/3

2 hours

SPECIMEN PAPER

Candidates answer on the spaces provided on the question paper.

Additional materials:

As listed in Instructions to Supervisors

Mathematical set

Electronic calculator

INSTRUCTIONS TO CANDIDATES

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Record all your observations and answers in the spaces provided on the question paper.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

FOR EXAMINER'S USE

1	
2	
TOTAL	

This specimen paper consists of 8 printed pages.

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1 In this experiment you will determine the spring constant, k , of a slinky spring.

- (a) (i)** Measure and record the mass, m , of the spring and the initial length of the spring, l_0 .

m _____ [1]

l_0 _____ [1]

- (ii)** Using data in **(a)(i)**, calculate, μ , mass per unit length of the spring,

$$\left[\mu = \frac{m}{l_0} \right].$$

[2]

- (iii)** Assemble the apparatus as shown in **Fig 1.1**.

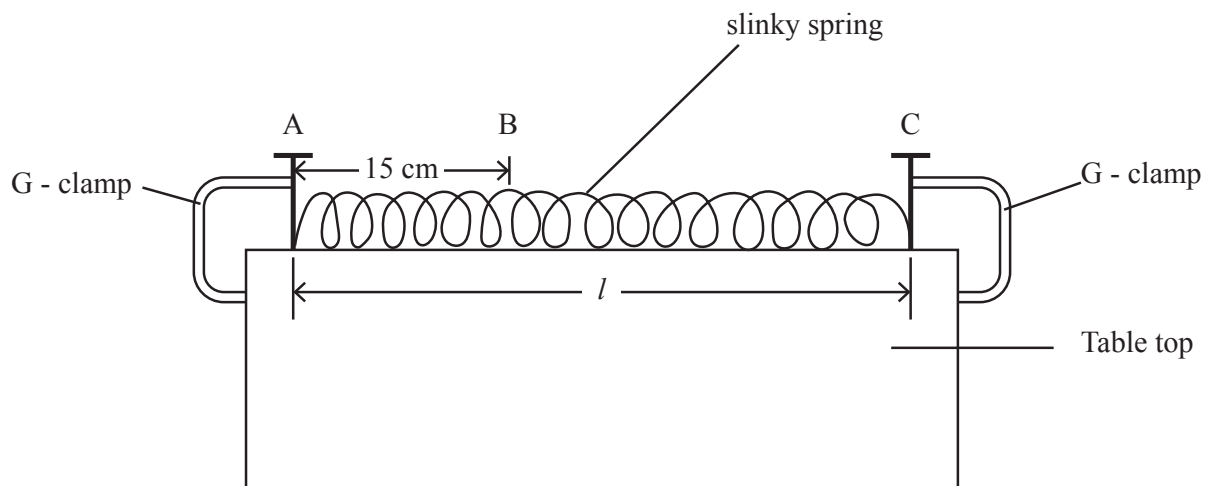


Fig 1.1

Measure and record l .

_____ [1]

- (b) (i) Pull the spring backwards from B to A.
- (ii) Simultaneously release the spring and start the stopwatch. Measure and record the time taken by the wave pulse to reach point C.
- (iii) Repeat (b)(i) and (b)(ii) to obtain two further sets of readings. Calculate the speed of the wave pulse using the formula, speed of wave pulse, $v = \frac{l}{t}$ where t is the time interval for the pulse to travel from A to C.

Table 1.1

	$t/$	$v/$
1		
2		
3		

[5]

- (vii) Calculate the average speed of the wave pulse.

[1]

- (c) Theory suggests that the average speed, v , of the pulse is given by

$$v = \sqrt{\frac{k(l-l_0)}{\mu}}$$

where k is the spring constant.

- (i) Calculate the value of k with units.

[3]

- (ii) State the significance of k .

[1]

- (d) (i) State any **one** assumption made in the experiment.

[1]

- (ii) State any **two** sources of error in the experiment.

[2]

- (iii) Explain how each of the errors in **d(ii)** may be reduced.

[2]

- 2 In this experiment you will investigate the moisture level of soil using an analogue sensor.
- (a) (i) You are provided with 5 soil samples, sample A, sample B, sample C, sample D and sample E.
- (ii) Set up the apparatus as shown in **Fig 2.1**.

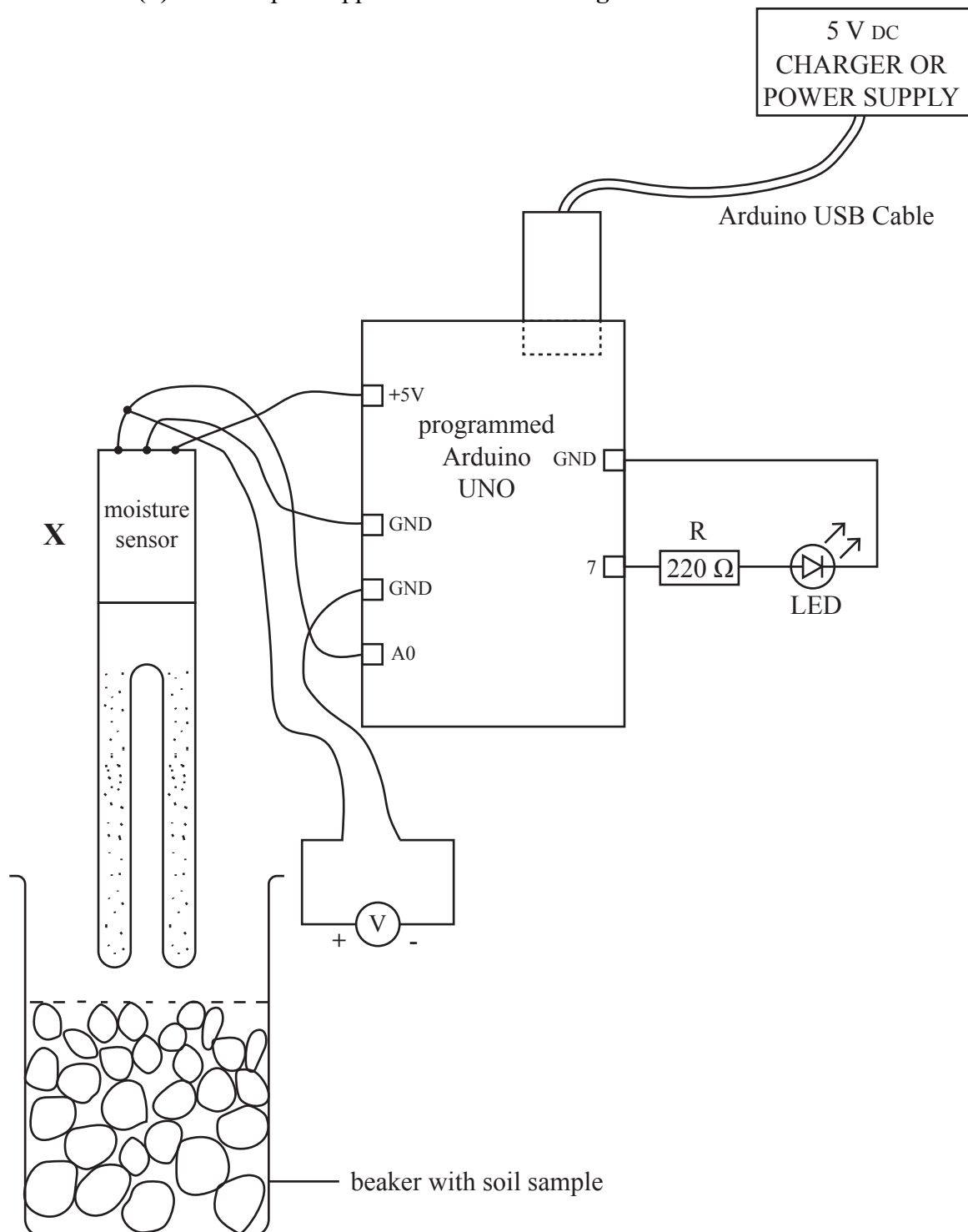


Fig 2.1

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[Turn over

- (b) (i) Switch on the programmed Arduino microprocessor.
- (ii) Insert the moisture sensor, X, into sample A. Measure and record the voltage reading in **Table 2.1**.
- (iii) Repeat (b)(ii) for samples B, C, D and E.

Table 2.1

soil samples	V/V
A	
B	
C	
D	
E	

[5]

- (c) (i) Write down the Arduino code to implement.

[6]

- (ii) Draw a bar graph of voltage against moisture content.

- (iii) Analyse the relationship between voltage and moisture content.

[4]

- (iv) Deduce the soil sample with highest and lowest moisture level.

Highest moisture level _____ [1]

Lowest moisture level _____ [1]

- (v) State any **two** precautions necessary in the experiment.

[2]

- (vi) Suggest any improvement that can be made in the experiment.

[1]